Youngjae Cho

ML Research Engineer

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youngjae-cho.github.io | GitHub | Google Scholar

I study robust and data-efficient learning from imperfect supervision: preference optimization, active learning, Bayesian adaptation, and reinforcement learning.

EDUCATION

KAIST | M.S., Industrial & Systems Engineering

Mar 2022 - Feb 2024

Advisor: Il-Chul Moon

KAIST | B.S., Industrial & Systems Engineering

Mar 2017 - Feb 2022

SELECTED PUBLICATIONS

NeurIPS 2026

Accepted

First author

[Paper]

Learning Where It Matters: Geometric Anchoring for Robust Preference Alignment

Youngjae Cho, Jongsuk Kim, Ji-Hoon Kim

Proposed a batch-conditioned adversarial anchor and **anchor-gap reweighting** for brittle preferences; improved AlpacaEval 2.0 LC win rate by **3.6 percentage points** over SimPO (Mistral-7B).

AAAI 2024

First author

[Paper] [Code]

Make Prompts Adaptable: Bayesian Modeling for Vision-Language Prompt Learning with Data-Dependent Prior

Youngjae Cho, HeeSun Bae, Seungjae Shin, YeoDong Youn, Weonyoung Joo, Il-Chul Moon

Developed Bayesian prompt adaptation with **data-dependent priors** and **Wasserstein gradient flow** for posterior inference, improving few-shot generalization under distribution shift.

ICML 2023

Co-first author

[Paper] [Code]

Sharpness-Aware Active Learning

Yoon-Yeong Kim*, **Youngjae Cho***, JoonHo Jang, Byeonghu Na, Yeongmin Kim, Kyungwoo Song, Wanmo Kang, Il-Chul Moon

Co-developed sharpness-aware acquisition through **pseudo-label-based perturbed-loss estimation**, with a theoretical optimization bound; evaluated classification, detection, and domain-adaptive segmentation.

arXiv 2026

Preprint

First author

[Paper]

Beyond Scalar IoU: Structured Verification from Rollout Groups for Video Temporal Grounding

Youngjae Cho, Won Young Jhoo, Jongsuk Kim

Introduced a group-conditioned verifier with an **exact decomposition into IoU and a covariance correction**; improved temporal grounding at every reported IoU threshold across five benchmarks.

* Equal contribution. Additional publications appear on page 2.

RESEARCH EXPERIENCE

Pyler
Oct 2025 - Present

ML Research Engineer

Preference learning. First-author GAPO research on local margin degradation under parameter perturbations; evaluated robustness to random and structured preference noise.

Structured verification. First-author SUTURE research on group-conditioned rewards; analyzed verifier structure and trained temporal-grounding policies with GRPO.

Multimodal post-training. Implemented rollout, preference loss, and training loop on Megatron-Bridge for Nemotron-Nano-12B-VL; designed SimPO preference data for video localization and content safety.

Auditable supervision. Recovered decision trees from 299K reasoning traces; improved content-safety macro F1 from **0.777 to 0.857**.

Aiv Co.
Mar 2024 - Oct 2025

ML Research Engineer

Generative modeling. Led background-aware diffusion research for scarce industrial anomalies; evaluated defect synthesis on MVTec-AD and LOCO and deployed it to improve production detector precision and recall.

KAIST
Mar 2022 - Feb 2024

Graduate Research

Statistical learning. First-author APP and co-first-author SAAL research combining Bayesian inference, optimization, and active learning; evaluated adaptation under distribution shift and acquisition across classification, detection, and segmentation.

ADDITIONAL PUBLICATIONS

arXiv 2024
Preprint
First author
[Paper]

Background-Aware Defect Generation for Robust Industrial Anomaly Detection
Youngjae Cho, Gwangyeol Kim, Sirojbek Safarov, Seongdeok Bang, Jaewoo Park

ICML 2022 Workshop
[Paper]

Improving Group-based Robustness and Calibration via Ordered Risk and Confidence Regularization
Seungjae Shin, Byeonghu Na, HeeSun Bae, JoonHo Jang, Hyemi Kim, Kyungwoo Song, **Youngjae Cho**, Il-Chul Moon

Workshop on Spurious Correlations, Invariance and Stability

IEEE SMC 2021
[Paper]

Predict Sequential Credit Card Delinquency with VaDE-Seq2Seq
Yeongmin Kim, **Youngjae Cho**, Hanbit Lee, Il-Chul Moon

ACADEMIC SERVICE

ICML 2026 | Gold Reviewer

NeurIPS 2026 | Reviewer

AWARDS

NVIDIA Nemotron Hackathon
Winner, Track B | 2026

Team winner in Track B (Domain-Specialized Model).